



Weka: A Free and Open Source Suite for Machine Learning and Deep Learning Algorithms

Weka or the Waikato Environment for Knowledge Analysis is a machine learning suite that is written in Java, having been developed by the University of Waikato in New Zealand. This article is an introduction to an effective ML tool.



Machine learning, deep learning and predictive analytics are the key domains of research in engineering, finance, economics, real-time imaging and many other fields. Researchers are working on different tools and technologies in these fields so that a higher degree of accuracy can be achieved.

The deep learning industry is very closely integrated with machine learning (ML), resulting in a higher degree of performance and accuracy with the minimum error rate. Figure 1 gives the predicted CAGR in revenues from deep learning in the USA, during the period 2014 to 2025.

Weka is a free and open source tool for machine learning and Big Data analytics (URL: <https://www.cs.waikato.ac.nz/ml/weka/>).

Table 1 lists the prominent tools and software libraries used for the machine learning and data science based implementations.

Although there are a number of software libraries being widely used, Weka is a powerful tool preferred by researchers and data scientists. It has a huge set of machine learning and data science based algorithms including Big Data analytics. Weka can be used with the command line interface as well as the graphical user interface (GUI) for the implementation of algorithms. Besides the inbuilt and pre-loaded packages in Weka, there are assorted extension packages that can be integrated for advanced applications.